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Worlds Beyond Earth: Planetary Atmospheres

Air on Other Worlds

Aira Astralis

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Introduction

Every world that matters wears a veil. The earth's is the blue sky, a hundred kilometers of gas that makes the planet livable, paints its sunsets, and carries its weather. Venus hides beneath a crushing carbon dioxide blanket hot enough to melt lead. Mars keeps only a wisp of its former self. Titan, a moon, holds an atmosphere thicker than the earth's, orange with organic haze. Jupiter and Saturn are almost entirely atmosphere, all the way down. These envelopes are planetary atmospheres, and this book is their story.

The atmosphere is the most changeable thing about a planet. Rock is a world's skeleton, slow to move and slower to change; gas is its breath, moving in hours, reacting in minutes, and evolving over the planet's whole life. The gas responds to everything: the sunlight it absorbs, the surface beneath it, the star's wind, the planet's own gravity and spin, the chemistry of its own molecules. This responsiveness is why atmospheres are the most informative layer to study and the hardest to understand, and it is why a planet's story is, more than half of it, an atmospheric story.

The atmosphere is also where a planet's history is written. A world's envelope is not a given; it is the outcome of a competition between sources and losses, and the competition's record, read in the present gas, tells the past: the earth's oxygen tells of life, Mars's isotopes tell of a lost world, Venus's heat tells of a runaway disaster. To read an atmosphere is to read a biography, and the tools of the reading, spectroscopy, probes, and models, are the tools this book sets out.

The book's plan is this. It begins with the fundamentals: where atmospheres come from, how they are held, and why each planet's envelope is the one it has. It covers the working machinery: the vertical structure, the radiative balance, the greenhouse effect, and the circulation that turns a static envelope into a weather system. It treats the chemistry, the clouds, and the escapes, the processes that make each atmosphere unique. It then tours the solar system's cases, from the terrestrial planets' contrasting fates to the giants' bottomless envelopes and Titan's cold mimicry of the earth. It ends with time: the climate changes, the evolution, and the exoplanet atmospheres, where the field's future lies.

The scale of the study deserves a closing note, since the atmospheres are the planets' most accessible parts. A world's interior is hidden beneath its envelope; its core is beneath its mantle; but the atmosphere is exposed, the layer the light crosses and the telescopes read, the first thing known about any planet and the last thing lost. The field's practical shape follows: the atmospheres' data, from the earth's four centuries of barometers to the exoplanets' transit spectra, are the planetary science's largest and longest archive, and the subject's richness is the archive's own richness, the atmospheres' study the planetary science's longest book.

One idea frames everything that follows: an atmosphere is a balance, and every balance can tip. The balance's terms are plain: gravity holding, heat releasing, sources feeding, losses draining, light flowing in and out, but the outcomes, from the earth's temperance to Venus's furnace, span the full range of a planet's possibility. The book's argument is that the balances are legible, that the reading is one of planetary science's central achievements, and that the atmospheres, once understood as the planets' most delicate and expressive organs, become the key to both our own world's fragility and the new worlds beyond the sun.

The final framing idea is the atmosphere's reach. The subject of this book is not a specialty of one planet but a general science, and the reader who learns the balances here will recognize them everywhere: in a brown dwarf's weather bands, in an exoplanet's evaporating tail, in the early earth's haze, and in the climate question the reader's own generation is asking. The atmospheres are the planets' faces, and learning to read them is learning to read the worlds.

The subject opens with an observation worth making before any chapter starts, one that reorganizes the whole tour. An atmosphere is not a thing a planet has but a thing a planet does, a standing balance between what the interior and the surface release, what the gravity holds, and what the wind and the sun remove, and every sky in this book works as a ledger being actively kept, with inflows and outflows the chapters will learn to read. The reason the subject exists as a discipline is the comparative one the conclusion will make explicit: the physics is shared, and the outcomes, across the planets this book visits, are wildly different.

Chapter 01: Origins and Retention

An atmosphere exists only because a planet both acquires gas and keeps it, and this chapter is the accounting:

where the envelopes come from, what holds them, and why the retention succeeds for some worlds and fails for others.

A note on the timing belongs here, since the channels' eras differ. The nebular capture is the earliest: the gas disk's lifetime is a few million years, and a planet that misses the window, one that grows slowly to a heavy core, arrives after the gas and gets no primary envelope. The outgassing is the adolescence: the planet's interior's heat engine, running on the accretion's and the radioactivity's energy, drives the volcanism that vents the volatiles, and the envelope's building is the first billion years's work. The delivery is the outer system's gift throughout: the comets' and the asteroids' flux, heavy in the era of the giant planets' migration, has declined since, and the present-day flux, the micrometeorites' and the interplanetary dust's steady rain, is the remnant of the channel that may have brought the earth's water. The sequence's consequences follow: the primary envelopes are old, the secondary envelopes are built, and the deliveries' records, the noble gases' and the isotopes', are the timing's testimony.

The acquisition has three channels. The first is the primordial capture: the youngest planets, forming in the gas-rich disk, can swallow the nebula's hydrogen and helium directly, and the giants did exactly this, capturing enormous envelopes while the disk lasted. The second is the volcanic outgassing: a solid planet's interior holds dissolved gas, and the volcanoes and vents release it, the steam and carbon dioxide and nitrogen that became the secondary envelopes of the terrestrial worlds. The third is the delivery: the comets and the volatile-rich asteroids, bombarding the young surfaces, brought ices and gases from the outer system, a contribution the earth's water and carbon remember.

The exosphere's threshold deserves a paragraph, since it is the retention's edge. The exobase, the height where the gas's collisions end and the molecular free flights begin, is the atmosphere's legal boundary in the escape's sense: below it, a fast molecule is knocked back; above it, a fast molecule is gone. The exobase's height, for the earth about five hundred kilometers, is set by the below's density and the above's temperature, and the hot thermosphere's temperature, controlled by the sun's ultraviolet, moves the exobase and the escape's rate with it: a solar storm's heating can raise the escape's flux by orders of magnitude for a day, a reminder that the retention's balance, on the geological scale a constant, on the daily scale is a weather's variable too.

The retention is gravity's work against heat's. A gas at temperature is a swarm of molecules, each moving with a speed set by the temperature and the molecule's mass, and the swarm's fast tail, the molecules moving fastest, is the part that escapes: any molecule whose speed approaches the planet's escape velocity can leave, and the leaving, molecule by molecule, is the atmosphere's loss. The balance's terms are the planet's mass, the strong gravity of a big world holding its gas, and the temperature, the cold outer system's worlds keeping ices the inner system cannot, and the molecule's mass, the heavy gases held where the light ones fly. The earth keeps nitrogen and oxygen but lost its hydrogen long ago; the moon, small and hot in daylight, keeps almost nothing; Jupiter keeps everything.

The polar wind deserves a paragraph, since the earth's escape is not only thermal. The field's own escape's channel, the polar wind's outflow, is the ionosphere's and the magnetosphere's coupling's own: the light ions, the hydrogen's and the helium's, flowing out along the polar field lines, the pressure's and the field's geometry's own engine, and the channel's contribution, the earth's steady hydrogen loss's majority, is the shield's own irony, the protector's field providing the escape's fastest road. The other channels' contributions, the photochemical escape's and the charge exchange's loads, complete the earth's total, and the planet's own escape deserves holding for its smallness: the earth's rates, the tons per second's hydrogen, are the geological nothing, the envelope's billion-year stability the numbers' own testimony.

The escape deserves the chapter's center, since it is the retention's negative and the evolution's engine. The thermal escape, the direct boiling away of the fastest molecules, is the simplest channel, and its threshold, the ratio of the escape velocity to the gas's typical speed, is the classic rule: a planet that cannot hold a gas at its temperature loses it. The non-thermal channels are subtler and, for the terrestrial worlds, dominant: the solar wind's ions, hitting the upper atmosphere, can knock molecules away, a process the magnetosphere's shield moderates; the photochemistry's products, given excess energy by the star's ultraviolet, can be flung out; and the sputtering, the ions' and neutrals' collisions in the exosphere, can carry heavier gas than the thermal channel allows. The escape's total rate, summed over the channels, is the atmosphere's clock, and the clock's reading is the chapter's point: the differences among the planets' envelopes are, to first order, the escape's different outcomes.

The balance's diagram is the classic figure of the field, and it deserves a paragraph. The planet's mass and the

temperature define a retention zone: the gases heavier than a threshold are held, the lighter ones lost, and the threshold moves with the planet. The giant planets sit deep in the zone, holding even hydrogen; the earth sits at the boundary, having kept the heavy gases and lost the light; Mars sits near the small-mass edge, keeping a thin remnant; and the moon and Mercury sit outside, their exospheres the merest traces, re-supplied and re-lost continuously. The diagram is a simplification, and the field's later chapters complicate it, but keep the plain fact: the envelopes we see are the survivors, and the survivors are selected by mass, temperature, and time.

The chapter's summary is the accounting's shape: the atmospheres are acquired by capture, outgassing, and delivery, and retained by gravity against the escape, with the balance's outcome set by the planet's mass, its distance from the star, and the gas's own weight. The corollary carries forward: no envelope is permanent, and the evolution, the slow change of the balance's terms over the planet's life, is the subject of the later chapters on climate and loss.

The origins file adds the timing ledger, because an atmosphere's birth certificate is partly a weather report of the early system. The primary envelopes, the hydrogens and helios of the Sun's original cloud, were caught or lost in the first splashes of formation, and the deadline was brutal: the young Sun's fierce output stripped the inner worlds to their socks within a few hundred million years, while the giants far out kept their primordial holdings by sheer mass and cold. The chapter's addition on timing is the field's standing correction of a common picture: the skies now overhead are not leftovers of the original cloud but second acts, exhaled, outgassed, and delivered, and the first acts survive only in the giant planets and in the strange noble-gas ratios the chemists keep as evidence.